

RAJESH K. JAT

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ML Systems Design	LLMs & GenAI Infrastructure	GPU Inference Optimization	Computer Vision at Scale	AI Platform & MLOps	Edge & Air-Gapped Deployments
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EXPERIENCE

- AI Systems Engineer**
WESEE — Naval R&D; Lab, Ministry of Defence, Government of India

New Delhi · Jul 2023 – Present

 - Designed and owned end-to-end **satellite imagery analytics platform** for maritime warship detection: architected the full pipeline from OBB-based detection model training, automated geolocation calculation, and vessel length measurement (Rasterio, QGIS) through to FastAPI backend and ReactJS operational dashboard; system is active across multiple naval commands.
 - Engineered a **real-time multi-camera surveillance analytics engine** with multi-object tracking, trajectory prediction, and behavioural pattern detection; achieved **<50 ms end-to-end inference latency** through TensorRT INT8 quantization, tiled inference, and a DeepStream pipeline on NVIDIA Jetson edge devices and CUDA Linux servers.
 - Architected and shipped a **classified offline GenAI platform** on air-gapped private cloud, comprising: an LLM-based RAG system for naval documentation retrieval and Q&A (LangChain, vLLM, ChromaDB, FastAPI, ReactJS); an automated report and presentation generator from institutional templates; reduced analyst documentation turnaround by an estimated **60%**.
 - Built a **multimodal document intelligence system** combining open-weight instruction-tuned LLMs with neural TTS for podcast-style audio summarisation of classified documents; system processes PDFs through a retrieval-augmented pipeline and generates broadcast-quality audio briefs, enabling eyes-free situational awareness for field personnel.
 - Designed and deployed an **agentic coding assistant** on self-hosted open-weight LLMs, integrating automated code generation, test scaffolding, and inline documentation into the engineering workflow; reduced boilerplate authoring time by **~40%** across the lab's engineering team.
 - Sole engineer responsible for **building the lab's complete AI/ML infrastructure from zero**: GPU cluster provisioning and administration, self-hosted GitLab and Jira, internal SMTP mail server, SeaweedFS distributed object storage, and all private-cloud model serving on fully air-gapped systems with zero external vendor dependency.
- AI Systems Engineer (Contract)**
Safespace Global Corporation

Remote · 2024

 - Trained and deployed a **custom Vision-Language Model** for a 911-inform alerting system across schools and public venues; built the complete smart-investigation pipeline covering real-time threat detection, evidence capture, and automated incident-report generation, reducing alert-to-report latency from minutes to **<15 seconds**.
 - Architected a **custom MLOps platform** covering model versioning, CI/CD pipelines, containerised deployments, automated rollback, and production drift monitoring, enabling zero-downtime model updates across multi-tenant environments.
- Computer Vision Engineer (Contract)**
Sponsorlytics

Remote · 2023 – 2024

 - Trained a **multi-stage logo detection and brand placement analysis system** on live sports broadcast footage; delivered automated analytics including frame-level visibility duration, on-screen position heat-maps, and sponsor ROI scoring, processing full-match footage **at broadcast frame rates in real time**.
- AI Engineering Intern**
Object Automation System Solutions Pvt. Ltd.

Chennai · Sep 2022 – Jan 2023

 - Developed a Flask-based collaborative filtering recommendation engine and administered a Linux VPS-hosted LMS maintaining 24/7 uptime for active learners.

TECHNICAL SKILLS

ML Systems	Object Detection & Tracking Architectures (OBB, multi-stage), Pose & Motion Estimation, Model Training & Fine-tuning, Geospatial & Satellite Imagery Analysis, OpenCV, FFmpeg, Rasterio, QGIS
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GenAI & LLM Infra	Open-Weight LLMs, Instruction-Tuned & Multimodal Models, RAG System Design (LangChain, LlamaIndex, ChromaDB), Agentic Workflows, Prompt Engineering, Neural TTS, Vector Databases, vLLM Serving
GPU & Inference	NVIDIA CUDA, TensorRT (INT8 / FP16), DeepStream, Slang, NVIDIA Jetson Edge Deployment, Quantization, Tiled Inference, Low-Latency Serving, Inference Benchmarking
Backend & APIs	FastAPI, Flask, REST APIs, Docker, Microservices Architecture, Air-Gapped & Private-Cloud Deployments, Python, C++
Frontend	ReactJS, Streamlit, JavaScript
Cloud & Infra	AWS (EC2, S3), Azure, SeaweedFS, Hadoop / HDFS, Self-Hosted GitLab, Jira, Linux Server Administration, Internal PKI & Mail Infrastructure
Data & Databases	PostgreSQL, MySQL, MongoDB, Vector Databases, SQL, R

EDUCATION

M.Tech — Artificial Intelligence & Data Science	2021 – 2023
IIIT Kota · MNIT Jaipur Campus	
■ Designed and deployed the institute's Hadoop distributed computing cluster , enabling faculty research on large-scale data processing workloads.	
B.Tech — Computer Science & Engineering	2017 – 2021
Rajasthan Technical University	

CERTIFICATIONS

AWS	AWS Fundamentals — Coursera
Cloud Native	Developing Cloud Native Applications — Coursera